

Jordan Dialpuri

Cambridge, United Kingdom | [linkedin.com/in/jordan-dialpuri/](https://www.linkedin.com/in/jordan-dialpuri/) | github.com/dialpuri

Postdoctoral research scientist at the MRC Laboratory of Molecular Biology, Cambridge. Focussing on developing next-generation deep learning methods for the structural biology software package Coot. Possessing a keen interest in creating innovative solutions to problems using machine learning and web-based applications.

EXPERIENCE

Postdoctoral Research Scientist

April 2026 – Present

MRC Laboratory of Molecular Biology

Cambridge

- Project focussing on improvement and modernisation of the popular structural biology software package Coot.
- Created novel large language model harness to leverage local open-source large language models to refactor the legacy codebase of Coot. Replaced requirement for approximately 10 years of tedious manual work with three weeks of automated code refactoring.
- Integrating deep learning methods developed during doctorate into Coot to advance model-building capabilities for nucleic acid and carbohydrates.
- Invited to teach Coot and model-building at international structural biology workshop.

Placement Student

June 2021 – September 2022

Afton Chemical

Bracknell

- Placement as part of fourth year of undergraduate degree, undertaking a research project in engine oil additive optimisation. Implementing novel machine learning models to predict the outcome of £150,000 industrial test.

EDUCATION

Doctor of Philosophy in Biological Chemistry

University of York

October 2022 – April 2026

- Title: Automated Model Building of Nucleic Acids and Carbohydrates Using Experimental Data and Deep Learning Models.
- Produced deep learning models which simulate the intuition of an expert crystallographer at understanding experimental data.
- Developed efficient automated software packages to produce accurate, high-quality macromolecular structures of nucleic acids and carbohydrates.
- Created innovative web application to allow easy and intuitive online carbohydrate structure validation.
- Invited to present machine learning work at many national and international conferences.
- Invited to teach structural biology at multiple structural biology workshops.

MChem in Chemistry

University of York

First Class Honours with Distinction (1*)

September 2018 - July 2022

SELECT PUBLICATIONS

NucleoFind: a deep-learning network for interpreting nucleic acid electron density

August 2024

- Produced three-dimensional U-Net model using TensorFlow to predict the positions of the phosphate group, sugar ring and nitrogenous base group in three-dimensional electron density.
- Transformed predicted positions into a three-dimensional model of nucleic acid automatically, reducing the need for manual modelling.
- Software produces more complete and accurate molecular models than existing methods, shortening nucleic acid structure solution from weeks or months to hours.
- Open source software written in Python and C++ and distributed on PyPi.
- Citation: **J. S. Dialpuri**, J. Agirre, K. D. Cowtan and P. S. Bond, *Nucleic Acids Research*, 2024, 52, e84

Other Select Publications

- **J. S. Dialpuri**, H. Bagdonas, L. C. Schofield, et al., *Acta Crystallogr F Struct Biol*, 2024, 80, 30–35.
- **J. S. Dialpuri**, H. Bagdonas, L. C. Schofield, et al., *Beilstein J. Org. Chem.*, 2024, 20, 931–939.
- **J. S. Dialpuri**, H. Bagdonas, M. Atanasova, et al., *Acta Crystallogr D Struct Biol*, 2023, 79, 462–472.

TECHNICAL SKILLS

Languages: Python, C++, JavaScript, R, Swift

Libraries: TensorFlow, PyTorch, React

Courses: Nvidia GPU programming

AWARDS

Margaret C. Etter Student Lecturer Award for exceptional scientific quality and potential in the field of crystallography

Kathleen Mary Stott Prize for excellence in scientific research